

# ISAAC ROSENBERG

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## EDUCATION

### • Erasmus Mundus Joint Master in Astrophysics

September 2025 – July 2027

*Master of Science in Astrophysics*

Italy, Serbia, France

- Two-year joint program across European universities focused on astrophysics and space science.

- **Current grade:** 29.7 / 30

- **Mobility Track:**

- \* Semester 1: University of Rome Tor Vergata (Italy)
- \* Semester 2: University of Belgrade (Serbia)
- \* Semester 3: Université Côte d'Azur (France)
- \* Semester 4: Institution to be determined

### • University of Toronto

September 2021 – May 2025

*Honours Bachelor's of Science*

Toronto, Canada

- Astronomy and Astrophysics Specialist, Mathematics Minor.

- **Grade:** 3.85 / 4.00

## RESEARCH EXPERIENCE

### The Impacts of AGN Feedback on Galaxy Cluster Environments

2024 - 2026

*Department of Astronomy, University of Toronto*

- Research project with Professor Renée Hložek and Dr. Martine Lokken.
- Ran hydrodynamic simulations with the 300 collaboration of galaxy clusters to understand how AGN feedback affects galaxy and cluster properties.
- Paper in preparation.

### Standard Siren Cosmology with Gravitational Waves

2024 - 2026

*Canadian Institute for Theoretical Astrophysics*

- Research project with Professor Maya Fishbach.
- Built a model that can simultaneously fit for the cosmological and astrophysical evolution of the binary black hole mass distribution.
- Received Summer Undergraduate Research Fellowship to continue this project over summer 2025.
- Paper in preparation.

### Summer Research, ATLAS Experiment

Summer 2024

*CERN, Switzerland*

- Worked under Professors Nikolina Ilic and Richard Teuscher on the ATLAS experiment at CERN for 12 weeks, in person.
- Investigated yearly spatial variations in jet distributions of the  $H \rightarrow ZZ$  particle decay.

## CONFERENCE, SCHOOLS AND ACADEMIC TALKS

C=CONFERENCE, S = SCHOOL, T=TALK

- [C.1] [STATSTRO](#), May 14, 2025, Toronto, Canada. Research talk on gravitational wave cosmology.
- [C.2] [CASCA](#), June 5, 2025, Halifax, Canada. Research poster and talk on galaxy clusters.
- [S.1] [EXPLORE](#), August 12, 2025, Toronto, Canada. Research talk on gravitational wave cosmology, winner of best talk award.
- [T.1] CMB - Large scale structure meeting at University of Cambridge, January 8, 2026, Cambridge, UK. Research talk on galaxy clusters.
- [T.2] Simulations meeting at University of Oxford, January 12, 2026, Oxford, UK. Research talk on galaxy clusters.
- [T.3] Cosmology meeting at Imperial College London, January 13, 2026, London, UK. Research talk on galaxy clusters.
- [T.4] FLATs talk series at University College London, January 13, 2026, UCL, UK. Research talk on galaxy clusters.
- [S.2] Master in Astrophysics and Space Science Winter School, February 23-26, 2026, Rome, Italy. Four day workshop with lectures and tutorials focussing on machine learning and data science applications to astrophysics.

Updated: Friday 19<sup>th</sup> June, 2026

HONORS AND AWARDS

|                                                                                                                                                                                   |                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| • <b>Winner of Best Talk Award at EXPLORE Summer School</b><br><i>York University</i>                                                                                             | 2025<br>[🏆]                 |
| • <b>Erasmus Mundus Joint Masters Scholarship</b><br><i>Erasmus MASS program</i>                                                                                                  | 2025 - 2027 €34,000         |
| • <b>Summer Undergraduate Research Fellowship</b><br><i>Canadian Institute for Theoretical Astrophysics</i><br>◦ Received \$9,500 to continue my research on gravitational waves. | Summer, 2025 \$9,500<br>[🌐] |
| • <b>Regents Admission Scholarship</b><br><i>Victoria College, University of Toronto</i>                                                                                          | 2021, \$5,000               |
| • <b>Regents In-course Scholarship I</b><br><i>Victoria College, University of Toronto</i>                                                                                        | 2022, \$1,000               |
| • <b>Prof. William Kingston and Dr. John Kingston Scholarship</b><br><i>Victoria College, University of Toronto</i>                                                               | 2023, \$1,000               |
| • <b>John Hamilton Watson Award for International Study</b><br><i>Victoria College, University of Toronto</i>                                                                     | 2024, \$3,000               |
| • <b>CIE International Experience Award</b><br><i>Centre for International Experience, University of Toronto</i>                                                                  | 2024, \$1,500               |
| • <b>Helen Mae Woodliffe Scholarship</b><br><i>Victoria College, University of Toronto</i>                                                                                        | 2024, \$1,000               |

TEACHING EXPERIENCE

|                                                                                                                                                                                                     |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Teaching Assistant</b><br><i>University of Toronto, Department of Mathematics &amp; Department of Astronomy</i>                                                                                  | 2023 - 2025 |
| • Teaching assistant for <a href="#">MAT186 (Calculus I)</a> , <a href="#">MAT187 (Calculus II)</a> , <a href="#">MAT188 (Linear Algebra)</a> , and <a href="#">AST251 (Life on Other Worlds)</a> . |             |
| • Led three tutorials per week, marked assignments, invigilated exams, and held office hours for university calculus, linear algebra, and astronomy courses with over 4,000 students combined.      |             |

RELEVANT COURSEWORK\* Graduate-level courses in italics

Astrophysics

- Stars and Planets, Galaxies and Cosmology, Introduction to Astrophysics, Astrophysics Laboratory, *Cosmology, Stellar Astrophysics, Radiative Processes, Active Galactic Nuclei, Plasma Spectroscopy, Computational Astrobiology, Interstellar Medium, General Relativity.*

Physics

- Classical Mechanics I & II, Electromagnetism I & II, Thermal Physics, Computational Physics, Physics Laboratory, *High Energy Physics, Quantum Mechanics.*

Mathematics and Computer Science

- Analysis, Multivariable Calculus, Linear Algebra I & II, Differential Equations, Introduction to Computer Programming, Neural Networks, *Complex Analysis, ODEs & PDEs.*

SKILLS

- **Programming Languages:** Python, C, C++, Perl, Java, MATLAB, ROOT
- **Hydrodynamic Simulations (and related tools):** GIZMO, GADGET, CAESAR, YT, DisPerSE
- **Python Packages:** Numpy, Scipy, Matplot, Pandas, Astropy, PyTorch, scikit-learn

PROFESSIONAL MEMBERSHIPS

- |                                                   |                         |
|---------------------------------------------------|-------------------------|
| • <a href="#">The Three Hundred Collaboration</a> | October, 2024 - Present |
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ADDITIONAL INFORMATION

**Languages:** English (native), French (intermediate), Russian (intermediate), German (beginner)  
**Interests:** World history, baseball, travel, chess, politics, music.